

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. in Computer Science and Engineering Branch : Computer Science and Engineering Semester :III Subject Name: Object Oriented Programming in Java Subject Code: BTCOC305-B Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following. (2X6)		12
A)	What are the primitive data types in Java? Write about type conversions.	1	
B)	What is a nested class? Differentiate between static nested classes and non-static nested classes.	2	
C)	What is a constructor? What is its requirement in programming? Explain with program.	1	
Q.2	Solve Any Two of the following. (2X6)		12
A)	Write a program to implement the Fibonacci series using for loop control structure.	3	
B)	Explain the various non-access modifiers used in java.	2	
C)	How to define a package? How to access, import a package? Explain with examples.	2	
Q. 3	Solve Any Two of the following. (2X6)		12
A)	How to declare and create array in JAVA? How to pass an array to method?	2	
B)	What is variable length argument, explain with program.	1	
C)	Write a Java program to sort a numeric array in ascending order	3	
Q.4	Solve Any Two of the following. (2X6)		12
A)	What are the benefits of inheritance? Explain the various forms of inheritance with suitable code segments.	2	
B)	Describe the uses of final and super keywords with respect to inheritance.	2	
C)	Design an interface called Shape with methods draw() and getArea(). Further design two classes called Circle and Rectangle that implements Shape interface to compute area of respective shapes. Use appropriate getter and setter methods. Write a java program for the same.	3	

Q. 5	Solve Any Two of the following. (2X6)		12
A)	How to handle multiple catch blocks for a nested try block? Explain with an example	2	
B)	Write a java program to create own exception for Negative Value Exception if the user enter negative value.	3	
C)	Explain pattern matching. What is the different meta characters used in pattern matching?	2	
	*** End ***		

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